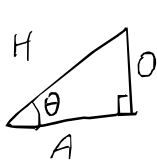


Basic Trigonometry

SOH - CAH - TOA



$$\sin \theta = \frac{O}{H}$$

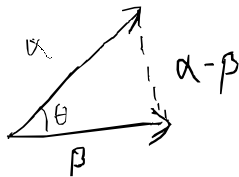
$$\cos \theta = \frac{A}{H}$$

$$\tan \theta = \frac{O}{A}$$

$$A^2 + O^2 = H^2$$

Imagine 2 vectors:

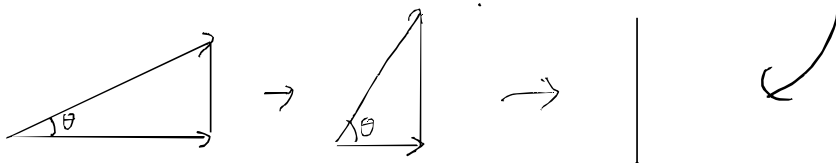
Dot Product



$$\alpha \cdot \beta = \|\alpha\| \|\beta\| \cos \theta$$

If $\theta = 90^\circ$:

$$\alpha \cdot \beta = \|\alpha\| \|\beta\| 0$$



$\therefore$ if the dot product equals 0,
we have orthogonal vectors, which implies linear $\perp$.

Cross Product



$$\text{Area} = h \times \|\beta\|$$

$$\alpha \times \beta = \|\alpha\| \sin \theta \|\beta\|$$

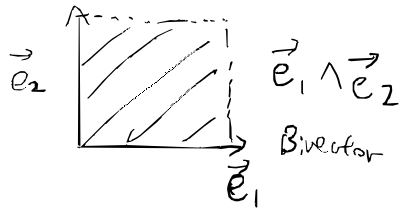
$$\alpha \times \beta = \|\alpha\| \|\beta\| \sin \theta$$

$$= (a_2 b_3 - a_3 b_2, a_3 b_1 - a_1 b_3, a_1 b_2 - a_2 b_1)$$

Wedge Product

(1) Antisymmetry

$$w \wedge n = -(n \wedge w)$$



(2) Nilpotency

$$v \wedge v = 0$$

(3) Bilinearity } Associativity

$$w \wedge (n + \theta) = (w \wedge n) + (w \wedge \theta)$$

Example

$$\alpha = 2dx + 3dy \quad \beta = 4dx - dy$$

$$\alpha \wedge \beta = (2dx + 3dy) \wedge (4dx - dy)$$

$$= (2dx \wedge 4dx) + (2dx \wedge -dy) + (3dy \wedge 4dx) + (3dy \wedge -dy)$$

$$= \underbrace{8(dx \wedge dx)}_{(2)} - \underbrace{2(dx \wedge dy)}_{(1)} + \underbrace{12(dy \wedge dx)}_{(2)} - \underbrace{3(dy \wedge dy)}_{(2)}$$

$$= 2(dy \wedge dx) + 12(dy \wedge dx)$$

$$= \boxed{14}(dy \wedge dx) \text{ or } -14(dx \wedge dy)$$

$\therefore$ Area is $\boxed{14}$ on the xy -plane,

clockwise from y to x or anti-clockwise from x to y .

Gram Matrix

$$\begin{aligned} \text{Cov}(x, y) &= \frac{1}{n} \sum (x_i - \bar{x})(y_i - \bar{y}) \\ &= \frac{\vec{a} \cdot \vec{b}}{n} \quad (\text{where } \vec{a} \text{ and } \vec{b} \text{ are demeaned}) \end{aligned}$$

$$G = X^T X = X X^T$$

If X is demeaned,

$$G = n \text{Cov}(X_i, X_j)$$

Naturally, if we have σ_i, σ_j we get correlation.

$$\text{Corr}(x, y) = \frac{\text{Cov}(x, y)}{\sigma_x \sigma_y}$$

Note that we diluted both the **first** and second moment. But in a market **crash**, one can assume all assets start displaying **no drift**.

Consider this: $a \cdot b = \|a\| \|b\| \cos \theta$

If we normalize our vectors, we get: $a \cdot b = \cos \theta$

$$\begin{aligned} a = \begin{pmatrix} 2 \\ 1 \\ 2 \end{pmatrix} \quad \|a\| &= \sqrt{9} = 3 & \frac{a}{\|a\|} &= \begin{pmatrix} 2/3 \\ 1/3 \\ 2/3 \end{pmatrix} & a^T a &= \frac{4}{9} + \frac{1}{9} + \frac{4}{9} \\ & & & & &= 1 // \end{aligned}$$

We are left with a unit vector.

$$G = \begin{bmatrix} 1 & & \\ & 1 & \\ & & 1 \end{bmatrix} \quad \text{is similar to a correlation matrix with the first central moment.}$$

Hence, we can measure market movements by calculating the volume.

Given that:

$$|a \wedge b|^2 = |a|^2 |b|^2 - (a \cdot b)^2$$

$$G = \begin{bmatrix} a \cdot a & a \cdot b \\ b \cdot a & b \cdot b \end{bmatrix}$$

$$\begin{aligned} \det(G) &= (a \cdot a)(b \cdot b) - (a \cdot b)^2 \\ &= |a|^2 |b|^2 - (a \cdot b)^2 \end{aligned}$$

$$\therefore |a \wedge b|^2 = \det(G)$$

$$\text{Volume} = \sqrt{\det(G)}$$